

Neighbourhood-Level Heat-Mortality Risk Across Canadian CMAs: A Case Time-Series Analysis

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Heat in Canada

- Heat is among the deadliest environmental exposures in Canada
 - 670 excess non-accidental deaths across 12 cities, 2000-2020
- Heat-mortality risk concentrates in specific neighbourhoods
- No DA-level heat-mortality estimates exist nationally.

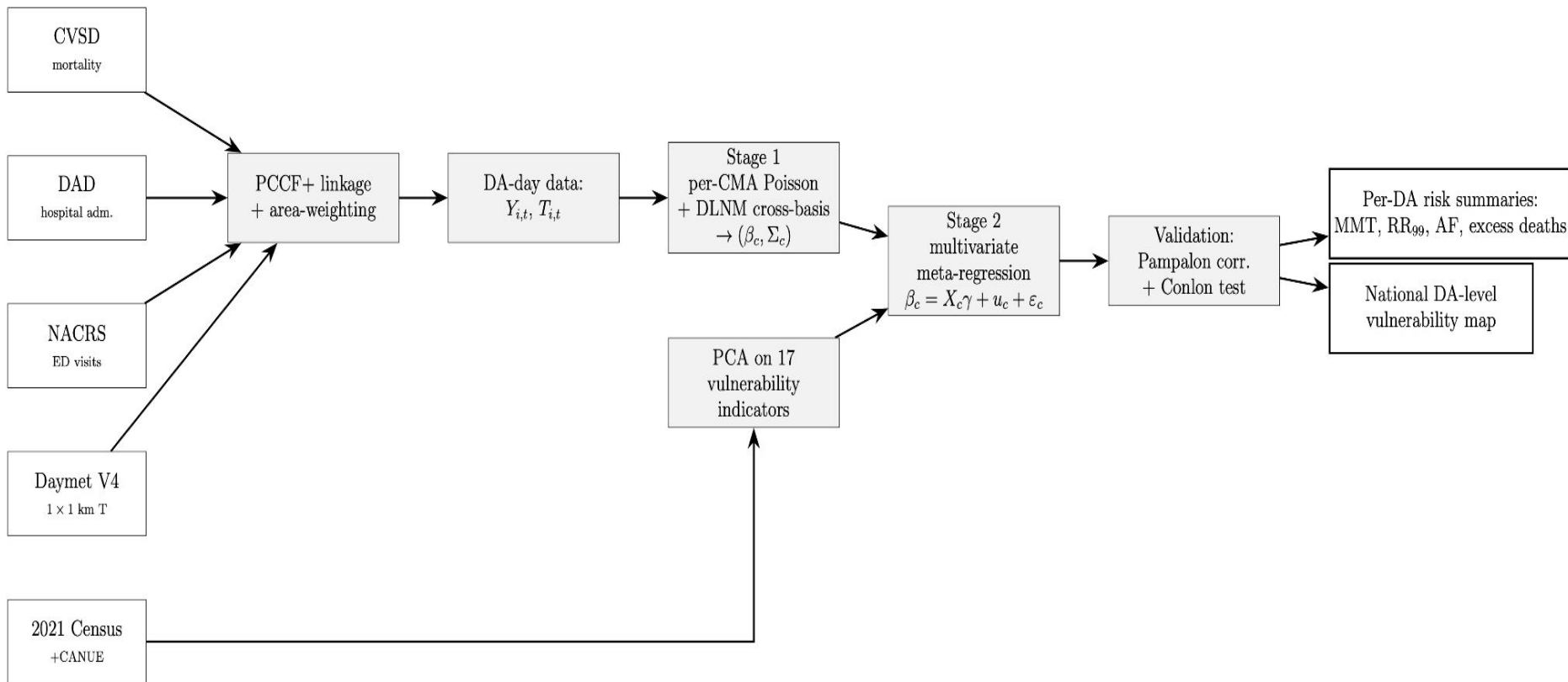


The cross-basis

$$cb(T, \ell) = \sum_{j=1}^4 \sum_{k=1}^4 \beta_{j,k} f_j(T) g_k(\ell)$$

- Heat-mortality risk depends jointly on temperature and lag
- The cross-basis is a tensor product of two splines: one in temperature and one in lag
- Gasparrini et al. used this framework across 34,753 small areas in England and Wales (Gasparrini et al., 2022)
 - we recombine this framework for Canada at the DA level across all 41 CMAs.

Analytical pipeline



Two-stage DLNM (Gasparrini et al.)

- Stage 1 estimates a per-CMA conditional Poisson at the DA level, with cross-basis $cb(T, \ell)$

$$\log E[Y_{i,t}] = \alpha_i + s(\text{time}_t) + \text{DOW}_t + cb(T_{i,t}, \ell; \beta_c)$$

- Stage 2 pools the 41 β_c vectors with CMA-level meta-predictors

$$\beta_c = X_c \gamma + u_c + \varepsilon_c$$

$Y_{i,t}$ = deaths in DA i on day t
 α_i = DA-specific baseline (absorbed via conditional Poisson)
 $cb(T, \ell; \beta_c)$ = cross-basis (next slide)
 X_c = CMA-level meta-predictors
 u_c, ε_c = between-CMA random effect, within-CMA sampling error

Assumptions

- We assume:
 - DA strata absorb all time-invariant confounders within each CMA
 - Daymet 1x1 km gridded temperature is a valid DA-level exposure proxy
 - a 21-day lag window captures the full mortality response to heat.

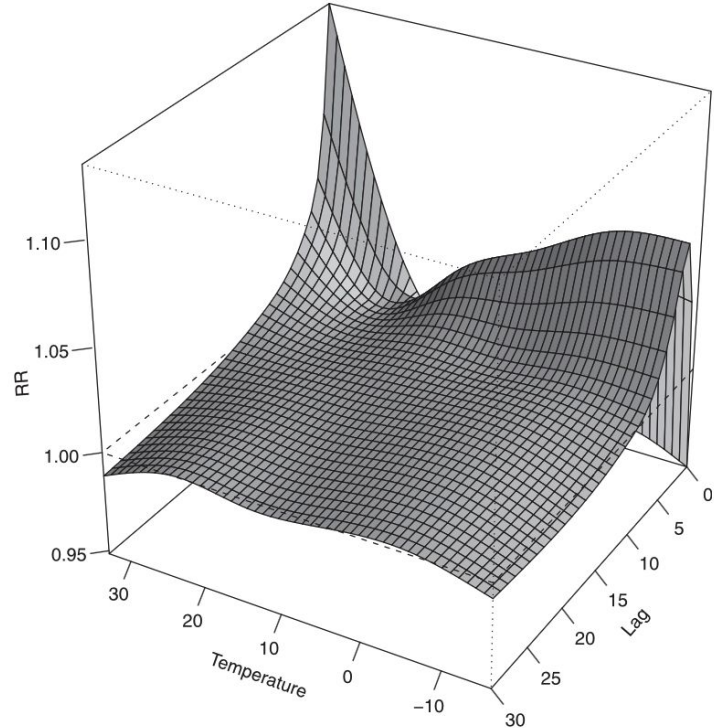
The cross-basis (visualized)

		Lag basis			
		0–1 d	2–7 d	8–14 d	15–21 d
Temperature basis	cold	$\beta_{1,1}$	$\beta_{1,2}$	$\beta_{1,3}$	$\beta_{1,4}$
	mild	$\beta_{2,1}$	$\beta_{2,2}$	$\beta_{2,3}$	$\beta_{2,4}$
	warm	$\beta_{3,1}$	$\beta_{3,2}$	$\beta_{3,3}$	$\beta_{3,4}$
	very hot	$\beta_{4,1}$	$\beta_{4,2}$	$\beta_{4,3}$	$\beta_{4,4}$

- $\beta_{4,1}$ captures acute mortality at extreme heat
- $\beta_{4,3}$ captures rebound at lag 8–14
- $\beta_{2,k}$ for mild temperature should be near zero.

$f_{j(T)}$ = natural cubic spline in temperature (knots at 10th, 75th, 90th percentile)
 $g_{k(l)}$ = natural cubic spline in lag (knots at lags 1, 4, 11)
16-dim β_6 per CMA

Lag-response surface



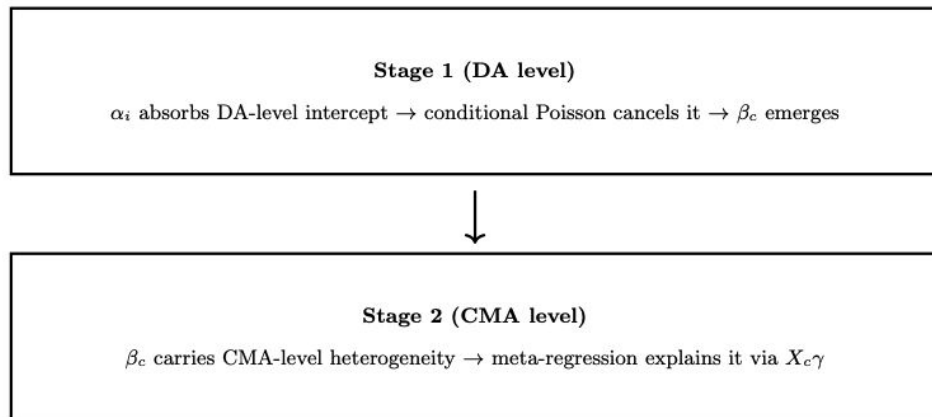
Gasparrini et al. (2010).

- Acute spike at high temperature, short lag
- Rebound below RR baseline around lag 15-20
- Cold effect rises gradually and persists across longer lags.

Vulnerability as Stage-2 meta-predictor

- To do this, we:
 - build a PCA-based vulnerability index from 17 Census and CANUE indicators
 - validate components against the Pampalon Material and Social Deprivation Index (Concordance within 0.5-0.85)
 - test components against Stage-1 DA-level risk estimates (Conlon et al.).

DA vs CMA stage



- Stage 1 absorbs DA-level confounding via α_i
- Stage 2 absorbs CMA-level heterogeneity in β_c
- Highest-resolution risk estimates without DA-specific models.

α_i = DA-specific baseline
 β_c = 16-dim cross-basis coefficient vector per CMA

Summary

- Prior work shows heat-mortality risk varies within and across Canadian cities
- Vulnerability dimensions could explain a quantifiable share of that variation
- The two-stage design transports Gasparrini's framework to all 41 Canadian CMAs.



Limitations

- We anticipate:
 - DA-level sparsity
 - PCA component instability across input specifications
 - NACRS coverage gaps outside Ontario, Alberta
 - Daymet exposure misclassification at sub-grid scales.

Conclusion

- We extend Gasparrini's case time-series design to all 41 Canadian CMAs at the dissemination-area level
- We expect to see vulnerability dimensions modifying the heat-mortality curve in quantifiable ways
- The resulting national vulnerability map establishes the spatial resolution required for neighbourhood-targeted heat adaptation in Canada.

Acknowledgments

- Thanks to my supervisors and committee for their guidance throughout
- Methodological framework adapted from Antonio Gasparrini and the MCC consortium
- This proposal reflects current understanding and will evolve as the thesis develops.

Thesis timeline

- StatCan & CIHI data delivery — Summer 2026
- Toronto pilot — July–August 2026
- National Stage-1 mortality fitting — Sept–Dec 2026
- Stage-2 meta-regression + vulnerability index — Sept–Dec 2026
- Morbidity extension — January–March 2027
- Manuscript and thesis writing — March–June 2027
- Defence target — Fall 2027.